

ATTACHMENT 30 – CONTRACTOR TRANSMITTAL (SUBMITTAL NUMBER 723.480-008 A) FROM BECHTEL / PARSONS BRINCKERHOFF (AUTHORIZED REPRESENTATIVE) TO MODERN CONTINENTAL COMPANY DATED MAY 27, 1999 REGARDING THE INSTALLATION PROCEDURE OF THE ADHESIVE ANCHORING SYSTEM AND THE USE OF SCREEN TUBES

(8 pages)



Central Artery/Tunnel

OFFICE COPY

CONTRACTOR TRANSMITTAL

TRANSMITTAL DATE: 05/12/99
TRANSMITTAL NO: CT-0118

CONTRACTOR: MODERN CONSTRUCTION

ATTENTION: REGENT STREET

CONTRACT NO: 99119 G0932

DOCUMENT TYPE

R - REPRODUCIBLE
P - PRINT
C - CATALOG OUT
E - EQUIPMENT MANUAL
S - SAMPLE
T - TECHNICAL DATA OR CALCULATIONS

REVIEW STATUS

1 - WORK MAY PROCEED.
2N - WORK MAY PROCEED SUBJECT TO THE CHANGES INDICATED. RESUBMITTAL NOT REQUIRED.
2R - WORK MAY NOT PROCEED ON INDICATED PART(S). RESUBMITTAL REQUIRED. WORK MAY PROCEED ON OTHER PART(S).
3 - WORK MAY NOT PROCEED. REVISE AND RESUBMIT.
4 - FOR INFORMATION ONLY.

DATE RECEIVED BY B/PS RE OFFICE: 5/12/99

DATE DUE BACK TO CONTRACTOR: 6/11/99

DATE SENT BACK TO CONTRACTOR: 5/27/99

RESIDENT ENGINEER: [REDACTED]

DATE: 5/27/99

ITEM	SUBMITTAL NO. (I.E., 120.010 - 011 - A)			DESCRIPTION	DOC DATE	DOC TYPE	REVIEW STATUS
	SPEC SECT	- SEQ#	- REV				
1	723.480	008	A	Precast Concrete Column System		T	3
2				Adhesive Anchor System			
3							
4							
5							
6							
7							

COMMENTS TO CONTRACTOR (USE ADDITIONAL SHEETS AS NECESSARY)

REVIEWER: [REDACTED] DATE: 5.26.99

WHITE - REVIEWER'S COPY GREEN - RESIDENT ENGINEER'S COPY YELLOW - INFORMATION COPY PINK - PROJECT ENGINEER COPY GOLD - CONTRACTOR'S COPY

CRE051 (8/96)

Precast Concrete Ceiling System Adhesive Anchorage System

INDEX STAMP

Contract No. 99119-C09B2	Paragraph 2.08
Specification Section (Spec-Seq-Rev) 723.480 - 008 A	

CENTRAL ARTERY / TUNNEL	
Permission To Proceed Does Not Constitute Acceptance Or Approval Of Design Details, Calculations, Analysis, Test Methods, Or Materials Developed Or Selected By The Contractor / Supplier And Does Not Relieve The Contractor / Supplier From Full Compliance With Contract Requirements.	
☐	1. Work May Proceed
☐	2N. Work May Proceed Subject To The Changes Indicated Resubmittal Not Required
☐	2R. Work May Not Proceed On Indicated Parts Resubmittal Required. Work May Proceed on Other Parts.
☒	3. Revise and Resubmit. Work May Not Proceed.
☐	4. For Information Only.
REVIEWED	
Gannett Fleming	5.26.99
Company	Date

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Please refer to comments marked herein.
A complete review was not made because fundamental
submittal requirements have not been complied with.
Comply w/ 1.04 K to confirm that the proposed
anchorage system will carry the design service loads
and provide evidence of compliance w/ Article 2.08
including the fire test.

Precast Concrete Ceiling Adhesive Anchoring System **Submittal No. 723.480 - 008 A**

The intent of this submittal is to provide all relevant Quality Control and material information for the provision and installation of the adhesive anchoring system required for the installation of the precast ceiling modules and other structural components, as described in Specification Section 723.480, Sections 1.04N and 2.08. The attached information from Power's Fasteners, Inc. (formally "The Rawlplug Company") shall be applicable to the anchors indicated in the contract documents, including but not limited to the following:

<u>Anchor Size</u>	<u>Sheet</u>	<u>Detail</u>
3/8" Diameter	S-279	Detail 2
REFER TO OTHER SPEC SECTIONS FOR ANCHOR REQMTS. 723.480 IS ONLY FOR THE PRECAST CONC. CEILING SYSTEM. SUBMIT OTHER ANCHORS SEPARATELY.	S-280	Section D
	S-366	Section D,F, Detail 6
	S-382	Detail 6
	S-391	Detail 1,3
	S-377	Detail 3A
1/2" Diameter		
5/8" Diameter	S-377	Detail 5
	S-381	Section A,B,B1
	S-382	Detail 5,7 (Section Z)
THESE DRAWINGS APPLY TO SECT. 723.480	S-387	Section A,B,C,D, Detail 2
	S-389	Section C
	S-279	DETAIL 2 & SECTION C.

Manufacturer's information on the above listed anchor sizes has been provided in Section 1 of the submittal package. Relevant information on all anchoring components, including but not limited to threaded anchor rods, nuts, washers, epoxy adhesive, dosage control screens, plugs, and locking devices has been included. Test results from the anchor test as described in Section 723.480, Section 2.08 B.1 have also been included for review.

Test specified materials.

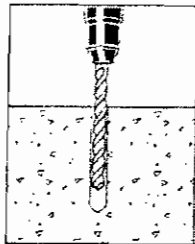
A Quality Control Program and installation procedure associated with the installation of the adhesive anchoring system has been included in Section 2.

1. Submit samples per 1.04.C.
2. Submit anchor system design calculations per 1.04 K.
3. Comply w/ Article 2.08. This Article contains fire test requirements, test loads, design service loads, etc.

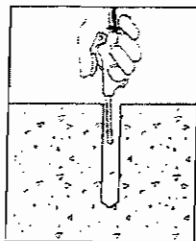
52.8 Power-Fast™ Installation Guidelines

52.8.1 Threaded Rods or Reinforcing Bars in Solid Base Materials

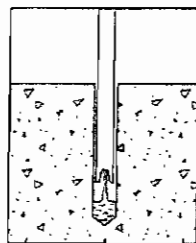
Drill a hole to the size and embedment required. The tolerances of the drill bit used should meet the requirements of ANSI Standard B212.15.



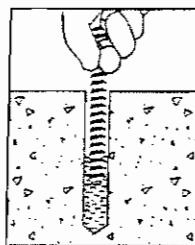
Blow the hole clean with compressed air, brush the hole, and blow it clean again. Holes should be clean and sound. They may be dry or damp, but should be free of standing water or frost. If using reinforcing bars, be sure the bar will fit into the drilled hole. If a larger hole is required, the diameter should be as close as possible to the diameter of the reinforcing bar.



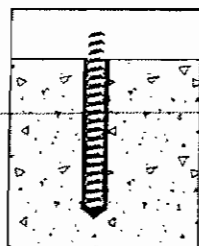
Be sure to properly balance the mixing nozzle prior to dispensing and when changing cartridges or sausages. Fill the hole approximately half way with epoxy starting from the rear of the hole. Slowly withdraw the static mixing nozzle as the hole fills to avoid creating air pockets within the hole.



Push the threaded rod or reinforcing bar into the hole while turning slightly to insure positive dispersion of the epoxy. Be sure the rod is fully seated at the bottom of the hole and that some epoxy has flowed from the top of the hole. The threaded rod or reinforcing bar used should be free of dirt, grease, oil, or other foreign material.

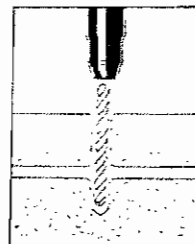


Allow the epoxy to cure for the specified time prior to applying any load. Do not disturb or load the anchor until it is fully cured.

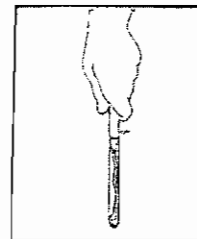


52.8.2 Threaded Rods in Hollow Base Materials

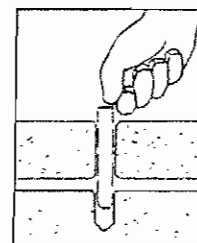
Drill a hole to the size and embedment for the required screen size. The tolerances of the drill bit used should meet the requirements of ANSI B212.15. Blow the hole clean with compressed air, brush the hole, and blow it clean again.



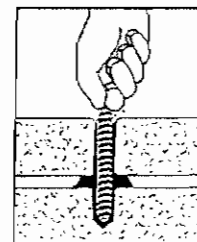
Properly balance the mixing nozzle prior to dispensing and when changing cartridges or sausages. Fill the screen tube with Power-Fast epoxy starting from the rear of the screen. The screen tube should be filled prior to inserting it in the hole.



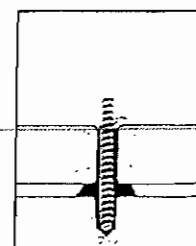
Insert the filled screen tube into the anchor hole until it is fully seated at the required embedment.



Push the threaded rod into the screen while turning slightly to insure positive dispersion of the epoxy. Be sure the rod is fully inserted to the end of the screen tube. The threaded rod or reinforcing bar used should be free of dirt, grease, oil, or other foreign material.



Allow the epoxy to cure for the specified time prior to applying any load. Do not disturb or load the anchor until it is fully cured.



Prior to use, read the box and carton label, Design Manual, Material Data Safety Sheet, and injection tool instructions.

Section 2

Installation Procedure & Quality Control Program

Adhesive Anchoring System

Power's Fasteners, Inc.

Section 2 - Installation Procedure & Quality Control Program **Adhesive Anchoring System**

It is the intent of Modern Continental Construction to perform the installation of the adhesive anchoring system in accordance with the included manufacturer's product information, and to utilize the accessories and equipment for epoxy application as supplied by the manufacturer to ensure proper installation of the adhesive anchor system. Modern Continental Construction shall make provisions to provide the proper temperature of adhesive materials and substrate during cold weather applications, per the manufacturer's recommendations.

The following detailed sequence of anchor installation is proposed in accordance with the Manufacturer's recommendations included in Section 1 of this submittal package.

1. Locations for the proposed structural element (clip, angle, etc.) shall be determined by survey and indicated on the substrate. The survey shall indicate the proper orientation, alignment and elevation (if applicable) of each structural element.
2. A template (or the actual structural element, if available) shall be used to determine the proposed location of each anchor required. The proposed location of these anchors shall be indicated on the substrate.
3. The holes shall be drilled in the proper locations. A template shall be used to ensure the drill and drill bit are properly oriented to the substrate while drilling the hole. A drill stop shall be used to ensure proper depth of drill hole. Drill hole diameter shall be determined in accordance with manufacturer's recommendations. MCC shall confirm that the employees performing the work are equipped with proper sizes of drill bits, anchors and accessories for the proposed installation.
4. The drill hole shall be brushed clean, free of drilling debris and dust. The anchor will be inserted into the hole to ensure proper sizing of the drill hole.
5. On floor applications, the drill hole shall be filled approximately one half full with the epoxy adhesive, starting from the rear of the hole.

On wall and ceiling applications, a screen tube shall be utilized to ensure proper application of the epoxy. Upon confirming the fit of the anchor in the drilled hole, the screen tube shall be filled with epoxy, starting from the rear of the screen tube. The screen tube shall be inserted into the drill hole, and fully seated at the required embedment depth. The hole shall be plugged with an anchor seal plug.

6. ~~The anchor shall be cleaned free of dirt, oil or foreign matter and inserted into the hole~~ (through the anchor seal plug in wall and ceiling applications), turning the anchor to ensure positive dispersement of the epoxy. The anchor shall be fully inserted to the proper embedment.

7. The anchor shall not be disturbed or loaded until after the minimum specified cure time for the epoxy, per the manufacturer's recommendations.